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Light+Wise

Persona

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Pain Point 1: Disconnecting the view of the dark winter sky and the rain road

Pain Point 2: Guilt as a citizen of an environmentally advanced city

"Our city is shouting 'Green Bellevue,'
I think it's contradictory to see the lights lit up in the empty parking lot all night.
That electricity is a waste."

Emily Chen (22, Female)

Occupation: Bellevue College student and downtown tech startup intern

Living conditions: Seattle/Bellevue region's unique winter (November to March) turns dark at 4:30 p.m., and drizzles continue to fall.

What's unique: On a day of class, you have to walk the wide and empty path from the campus building to the parking lot, and walk about 15 minutes from the company to the bus stop when you return home.

Washington state has low road reflectivity due to heavy rain.
With the existing streetlights, which are infrequent
There's a dark section like "Black Puddle"
It creates extreme anxiety for pedestrians.

Washington state residents are very environmentally conscious.
I saw a streetlight with 100% brightness at 3am when no one was there
Rather than being safe, it's a waste of energy and light pollution
I'm aware of it and I'm uncomfortable.

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Pain Point 1: High Cost Maintenance Structure

"In the United States, sending out a truck and a worker to fix a street lamp breaks hundreds of dollars.

But when you go there, it's often a simple sensor error.

If the office could know this in advance, it would save tens of thousands of dollars a year..."

Pain Point 2: Difficulty in proving policies due to lack of data

"The board presses every year to 'reduce carbon emissions'.

However, if you turn off the lights carelessly, there will be a security complaint.

"It's crazy because we don't have data to prove where and how much it's safe.

Rather than simply reporting that we've replaced it with an LED, we need data that can be quantitatively explained. Current systems have limitations in collecting such data automatically and in detail."

Michael Davis (52, Male)

Occupation: Director of Facilities Management

What's unique: Washington state's **strict 'energy code' and carbon emissions reduction targets** must be met. At the same time, **a large area must be managed with a limited budget**. In particular, the cost of sending maintenance personnel is very high.

Objective: Report 'reduced carbon emissions' performance to the City Council.

Unlike Korea, labor and business trips are very expensive in the United States.

The way to respond after receiving a citizen's report is Cost-effectiveness is the worst.

Washington is aggressively pursuing green policies, especially for energy efficiency and carbon reduction projects, data that quantitatively illustrates the expected savings and performance is an important factor in securing budgets and approving projects.

Hypotheses

1. Converting the existing always-on streetlight system to a traffic-based smart dimming system will reduce energy consumption by 30-60% without compromising public safety.
2. Even if the method of lowering the illuminance and immediately increasing the brightness only when motion is detected is applied in normal times, citizens' perceived safety will be maintained or their satisfaction will increase due to reduced light pollution.
3. If the status of individual streetlights can be monitored and controlled on the cloud dashboard in real time, managers will be able to increase maintenance efficiency and utilize traffic pattern data for urban planning without on-site inspection.

MVP

Configuring a Small IoT Streetlamp Sensor Network (Small Fleet)
Building LoRaWAN communication-based backend and web dashboards

For Citizen : "On a rainy winter night, a reliable partner who recognizes me and brightens the way"

For Manager : "A smart conductor who coordinates the entire city efficiently in the office"

MVP Plan

MVP Overview and Objectives (Pilot Phase)

Technical demonstration of key hypotheses with small sensor networks and integrated control systems

Core Configuration

[Device] Small IoT streetlight sensor unit (small unit)

[Backend] Serverless backend for sensor data collection and storage

[Web Dashboard] Real-time health monitoring, basic graphs, energy savings display

[Control] Simple Control Policy Settings and Manual Control (Manual Override)

System Architecture

[IoT Edge (Unit)] ESP32 + mmWave + LoRa | Features: Local Safety Logic (Independent Operations Without Internet)

[Connectivity (Gateway)] Raspberry Pi + Laura WAN | Features: Cost-Effective

[Backend] AWS IoT Core + Lambda + Dynamo DB/PostgreSQL

[Front End] React-based Web Dashboard

Functional Specifications

Implement specific functionality for hypothesis resolution

real-time control (Map/Grid View)

Fault Cause Remote Diagnosis
→ Prevent Truck Roll

Data Analysis (Analytics)

pedestrian detection heatmap,
Cost Reduction (ROI) Dashboard
→ Policy Proof

Manual Control (Manual Override)

Force ON/OFF,
Immediate Response to Emergency

Safe Mode (Fail Safe)

Standalone drive in case of network loss,
Maintain safe
brightness in case of sensor failure

Improve existing inefficient lighting methods to save energy and ensure citizens' safety

Hypothesis Validation

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Track A _ Citizen / Pedestrian

Night Walking Safety Verification

Objective: Demonstrate correlation between illuminance imbalance and avoidance behavior (quantitative survey)

[Research design]

Target: Campus and urban floating population (Emily Chen persona)

Methods: Offline panel (A/B test) and mobile QR questionnaire

Strategy: mapping the physical environment (status of street lights) and psychological data

Psychological indicators: Sensory anxiety index (1-5 points), lighting satisfaction

Behavioral Indicators (Key): Avoidance (Yes/No),

Avoiding frequency (two to four times a week, almost every time, etc.),

Priorities (whether to choose a bright path even if you return home late)

Track B _ Manager

Validation of maintenance efficiency

Objective: Identify the inefficiency and data needs of reactive management (qualitative research)

[Research design]

Target: Facilities management team leader, local government practitioner (Michael Davis persona)

Method: 1:1 in-depth interview

Action Strategies for Hypothesis Verification

1. Building Prototypes

Venue and destination: Some areas on the campus of Bellevue College

Scale: Install smaller, physical hardware and operate fewer light zones

Network Configuration: On-campus rooftops or utility poles

Install one LoRaWAN gateway

2. Track A Proof Plan: Civil Safety

Verification Indicator: Safety Brightness Maintenance Rate

During the time the motion is detected, the illumination is set

Measurement of the percentage maintained

above the 'safe brightness threshold'

Objective: React 100% instantly when

motion is detected to achieve safe brightness

Sensor Log Analysis: Delta Record

for Sensing Time vs. Illumination Rise Time

Fail-Safe Test: In case of loss of communication or sensor failure

Test to automatically return to 'default brightness'

3. Track B Proof Plan: Management Efficiency

Verification Indicator 1: Energy Saving Rate

Comparison group: Estimated amount of power

when left at 100% brightness overnight as usual

Experimental group: Actual amount of power measured after application of LightWise

Analysis: Calculate the difference between the two data

Calculate specific kWh savings and cost conversion

Verification Indicator 2: System uptime and reliability

Measure the ratio of the time the system was operating normally to the total operating time

Verify that the dashboard reflects the failure of individual streetlights in real time

MVP Development Process

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Phase 1 (Edge)

ESP32 & mmWave Sensor Integration, Independent Drive Firmware Development



Phase 2 (Network)

Building a LoRaWAN and Raspberry Pi gateway communication network



Phase 3 (Cloud)

Implementation of a serverless architecture based on AWS IoT Core and AWS Lambda



Phase 4 (Web)

Implement React-based control dashboard and real-time polling



Phase 5 (Pilot)

Installation and Operation of Prototypes in Bellevue College Campus

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SMART LIGHT SYSTEM